



# GeoDecarb: A GIS & Remote Sensing Framework for Steel Industry Decarbonisation

**Spatial Intelligence for Net-Zero Steel**



## **Problem:**

- Steel contributes ~8% of global CO<sub>2</sub> emissions.
- Current methods are energy-intensive, with limited spatial insight.

## **Solution:**

- A GIS & Remote Sensing-based framework to:
  - Identify emission hotspots
  - Map renewable integration zones
  - Optimise logistics and waste recycling
  - Support data-driven decarbonisation strategies



# WHY GIS & REMOTE SENSING?

- **Remote Sensing:** Detect land use, track emissions, assess renewable potential.
- **GIS:** Analyse spatial patterns, map hotspots, model scenarios.
- **Integration:** Combines real-time satellite intelligence with spatial analytics → precise, location-based decarbonisation planning.

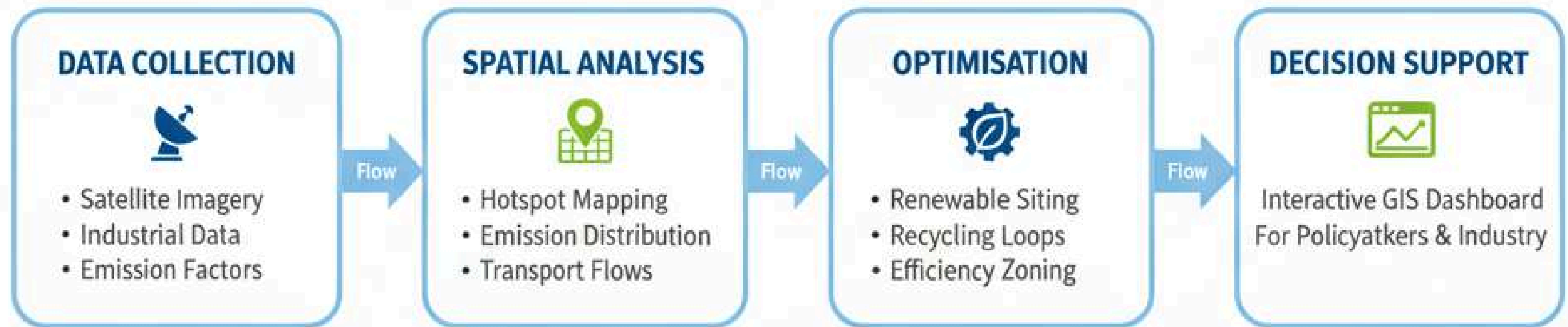


# FRAMEWORK OVERVIEW

- **Data Collection** – Satellite imagery, industrial data, emission factors.
- **Spatial Analysis** – Hotspot mapping, emission distribution, transport flows.
- **Optimisation** – Renewable siting, recycling loops, efficiency zoning.
- **Decision Support** – Interactive GIS dashboard for policymakers & industry.



# FRAMEWORK OVERVIEW





# NOVELTY & INNOVATION

- **Novelty**: First framework uniting industrial decarbonisation with spatial intelligence.
- **Innovation**: Focuses not just on what technologies to use, but where to deploy them most effectively.
- **Differentiation**: Goes beyond process engineering → adds geographic decision-making to accelerate decarbonisation.



# COMPETITORS AND CHALLENGES

## Existing solutions :

- Carbon monitoring dashboards
- Energy consulting models
- Process-based efficiency tools

## Challenges :

- Limited open industrial data
- Adoption barriers in heavy industry
- Scaling framework to prototype

**Your advantage :** Unique GIS + Remote Sensing integration → location-based precision.



# **TECH STACK & BUSINESS MODEL**

## **Tech stack:**

- Remote Sensing: Sentinel, Landsat, MODIS
- GIS: QGIS/ArcGIS, Google Earth Engine
- Data: OpenStreetMap, industry databases

## **Business model:**

- B2B SaaS/consulting platform for steel plants & regulators
- Provides emission maps, renewable siting, and logistics optimisation
- Scalable to cement, mining, and other heavy industries



# OUTCOMES AND STRATEGIES

## Expected Outcomes:

- Reduced carbon emissions
- Optimised energy & resource use
- Cost-effective decarbonisation pathways

## Strategy:

- Pilot with regional steel clusters
- Scale nationally → expand globally
- Extend to wider heavy industry sector



# CONCLUSION

- GIS & Remote Sensing provide a new lens for decarbonising steel.
- The framework delivers where + how to act.
- Offers a scalable, data-driven pathway to Net Zero.